

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
```

```
df=pd.read_csv("customer_retail.csv")
```

```
print(df.head())
```

	InvoiceNo	StockCode	...	CustomerID	Country
0	536365	85123A	...	17850.0	United Kingdom
1	536365	71053	...	17850.0	United Kingdom
2	536365	84406B	...	17850.0	United Kingdom
3	536365	84029G	...	17850.0	United Kingdom
4	536365	84029E	...	17850.0	United Kingdom

```
[5 rows x 8 columns]
```

```
df=df.dropna()
```

```
df=df[['Quantity', 'UnitPrice', 'Country']]
```

```
encoder=LabelEncoder()
df['Country']=encoder.fit_transform(df['Country'])
```

```
plt.figure(figsize=(8,5))
plt.scatter(df['Quantity'],df['UnitPrice'])
plt.xlabel('Quantity')
plt.ylabel('UnitPrice')
plt.title('Scatter Plot of Quantity vs UnitPrice')
plt.show
```

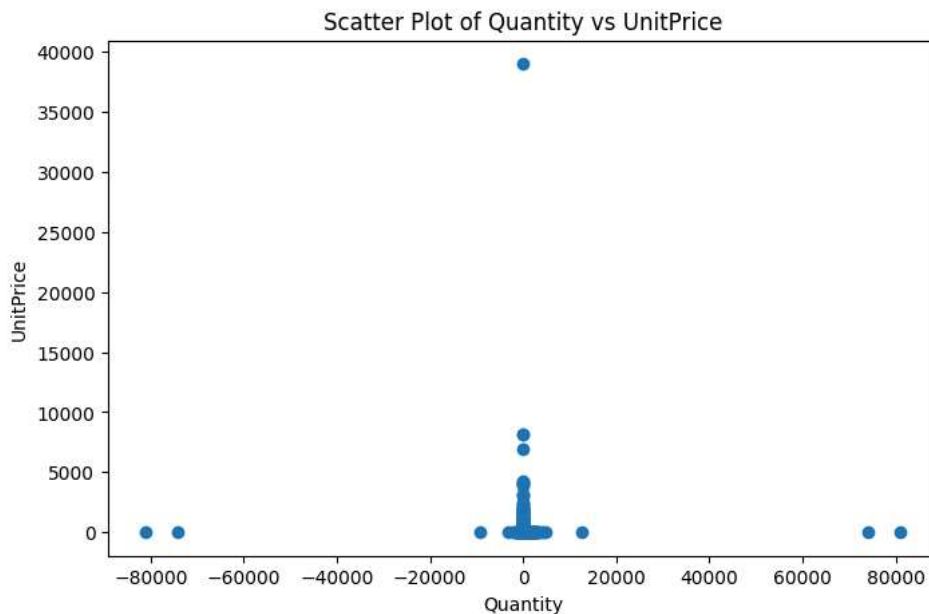
```
matplotlib.pyplot.show
def show(*args, **kwargs) -> None
```

Display all open figures.

Parameters

block : bool, optional

Whether to wait for all figures to be closed before returning.



```

x=df[['Quantity','UnitPrice']]
y=df['Country']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)

print("Logistics Regression")
log_model=LogisticRegression()
log_model.fit(x_train,y_train)
y_pred=log_model.predict(x_test)
log_reg_accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",log_reg_accuracy)
cm=confusion_matrix(y_test,y_pred)
print("Confusion Matrix:")
print(cm)

print("Decision Tree")
dt_model=DecisionTreeClassifier()
dt_model.fit(x_train,y_train)
y_pred=dt_model.predict(x_test)
dt_accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",dt_accuracy)
print("Confusion Matrix:")
cm=confusion_matrix(y_test,y_pred)
print(cm)

print("KNN")
knn_model=KNeighborsClassifier()
knn_model.fit(x_train,y_train)
y_pred=knn_model.predict(x_test)
knn_accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",knn_accuracy)
print("Confusion Matrix:")
cm=confusion_matrix(y_test,y_pred)
print(cm)

```

Logistics Regression

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1): STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

n_iter_i = _check_optimize_result(

Accuracy: 0.8897451023769142

Confusion Matrix:

```

[[ 0  0  0 ...  0 229  0]
 [ 0  0  0 ...  0 78  0]
 [ 0  0  0 ...  0 5  0]
 ...
 [ 0  0  0 ...  0 13  0]
 [ 7  0  0 ...  0 72395  0]
 [ 0  0  0 ...  0 43  0]]

```

Decision Tree

Accuracy: 0.8903104490819261

Confusion Matrix:

```

[[ 13  0  0 ...  0 197  0]
 [ 0  0  0 ...  0 75  0]
 [ 0  0  0 ...  0 5  0]
 ...
 [ 0  0  0 ...  0 13  0]
 [ 21  0  0 ...  1 72219  0]
 [ 0  0  0 ...  0 43  0]]

```

KNN

Accuracy: 0.8842268269301674

Confusion Matrix:

```

[[ 25  0  0 ...  0 181  0]
 [ 0  0  0 ...  0 75  0]
 [ 0  0  0 ...  0 5  0]
 ...
 [ 0  0  0 ...  0 12  0]
 [ 41 2  0 ...  0 71636  0]
 [ 0  0  0 ...  0 43  0]]

```

```

models=[
    'logistic regression',
    'decision tree',
    'KNN'
]

```

```
accuracies=[  
    accuracy,  
    accuracy,  
    accuracy  
]
```

```
plt.figure(figsize=(8,5))  
plt.bar(models,accuracies)  
plt.xlabel('Models')  
plt.ylabel('Accuracy')  
plt.title('Model Accuracy Comparison')  
plt.show()
```

